

1,181 Crystal Structures (ASE Atoms)

elastic_tensor_2015 · 8 target properties



5-Fold Cross-Validation

train = 500 | test = 681



ORB Featurizer (orbff-v2)

Forward pass → decoder hook → node embeddings

(N atoms × 256) → Mean pool → (1 × 256)



PCA Dimensionality Reduction

Fit on train · $d \in \{10, 25, 50\}$



BoTorch SingleTaskGP (×8 properties)

RBF Kernel + ARD lengthscales

Marginal Log-Likelihood Optimisation → Fitted Posterior



Posterior Predictions

Mean $\hat{y}$ (681×1) + Variance σ^2 (681×1)



Evaluation Metrics

R^2 · RMSE · Spearman ρ (per property, per fold)



Data / Input



Processing



Surrogate Model



Output / Evaluation